svpvolt

Anything happened about the product, please feel free to contact our customer service team at any time.

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Made in China

svpvolt

12.8V 50Ah

User Manual

Low-Temp Bluetooth



Please note: Part of the instructions is applicable only to the Bluetooth version.

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THINGS TO KNOW

BEFORE USING

When you open the package for the first time, please make sure the product works properly.

Do not install the product in harsh environments such as those that are damp, greasy, flammable, explosive, or prone to heavy dust accumulation.

When using the battery, please be careful to avoid metal or conductive objects touching the positive and negative poles of the battery at the same time; otherwise, it may cause a short circuit.

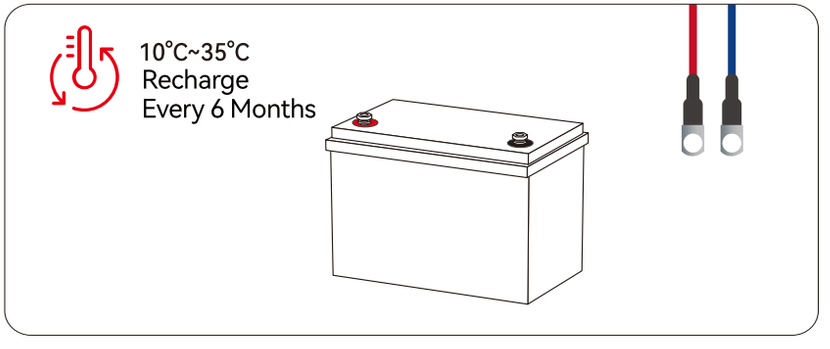
Install the battery upright with the post bolt facing up, and do not mount it upside down.

Tightly screw in the post bolts. Loose battery terminals will cause heat to build up, resulting in damage to the battery.

This battery is not intended to be used to start any devices; please DO NOT use it as a starting battery.

The battery can be operated at temperatures from -30°C to 60°C (-22°F to 140°F). A temperature between 10°C and 35°C (50°F to 95°F) is ideal for long-term storage. Store the battery in a fireproof container and away from children.

For a longer-lasting product, it is best to store the battery at a 60% charge level and recharge it every three months if it will not be used for an extended period.



PRODUCT OVERVIEW

12.8V 50AH BATTERY

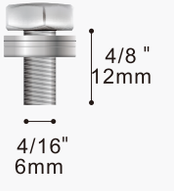
- Operating Voltage: 12.8V
- Charging Voltage: 14.6V
- Max. Continuous Charging Current: 50A
- Max. Continuous Discharge Current: 50A
- Max. Continuous Load Power: 640W



ADDITIONAL COMPONENTS

M6- 4/8" (12MM) TERMINAL BOLTS

The terminal bolts are used to secure multiple cable lugs to a single battery terminal. The bolts can be replaced with M6 bolts of other lengths based on actual needs.



INSULATING CAPS FOR BOLTS

24/7 MONITORING VIA

JBD BMS APP(Bluetooth version)

If your battery is not a Bluetooth model, please disregard this section.

X svpvolt intelligent Bluetooth LiFePO4 battery, integrated with Bluetooth 5.0, allows mobile phone viewing of battery status as well as real-time tracking and management with ease.

Step①

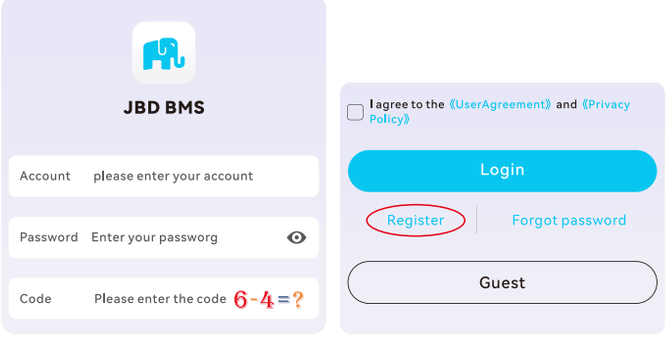
Scan the QR code with your browser, download the JBD BMS app, and register your account.

Download and install the App

JBD BMS

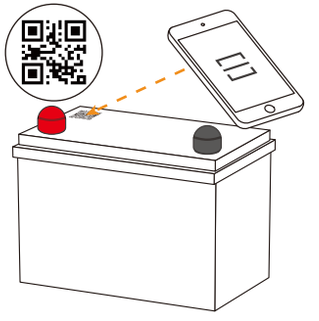
Step②

Open the app and register a username. Fill in the registered username and password to log in.

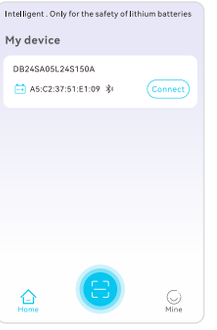


Step③

Pair the battery with the BMS APP by scanning QR code or searching Bluetooth and effortlessly keep track of the battery's real-time status.

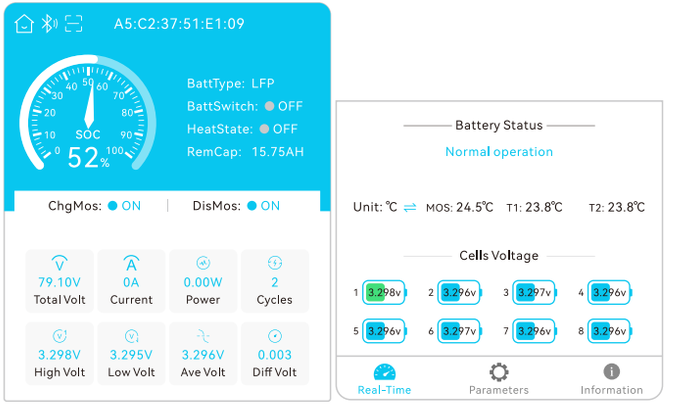


or



Step④

View product information on the app. On the main interface, users can directly see battery data such as battery level, voltage, current, temperature, and cell voltage.



BATTERY

PARAMETERS

Item	Parameter
Cell Type	LiFePO4
Combination Method	4S1P
Rated Capacity	50Ah (25A discharge)
Rated Energy	640Wh
Rated Voltage	12.8V
Battery Internal Resistance	IR≤26m Ω
Charging Voltage	14.6V
Discharge Cut-off Voltage	10V
Standard Charging Current / Interface	25A/M6 internal thread
Standard Discharge Current / Interface	25A/M6 internal thread
Maximum Charging Current / Interface	50A/M6 internal thread
Maximum Discharge Current / Interface	50A/M6 internal thread
Cycle life	≥4000 Cycles
Operating Temperature	Charging: 0 ℃~55 ℃ Discharge: -20 ℃~60 ℃

Item	Parameter
Storage Temperature	12month : 15℃~30℃
	8month : 10℃~40℃
	6month : 5℃~45℃
Net Weight	5.15kg
Storage Relative Humidity	65±20%
Factory Charge Capacity	SOC 38±5%
Product Dimensions(L*W*H)	197 * 134 * 176mm
Electrical Performance	Charging end pressure difference ≤ 200mV Discharge end pressure ≤ 300mV ≥ 50Ah (25A discharge at 25 ℃) ≥ 49.5Ah (50A discharge at 25 ℃) Test method: a) Charge with a constant current and voltage of 14.6V/25A, with a cut-off current of 3A or the highest single cell cut-off of 3.65V, and record the voltage difference at the end of charging; b) Let it stand for 30 minutes; c) Discharge at 50A until the total voltage reaches 10V, or until the lowest individual voltage reaches 2.3V. Record the voltage difference at the end of the discharge and the discharge capacity; d) Let it stand for 30 minutes; e) Charge at a constant current of 25A until 40% SOC is reached.

HOW TO ESTIMATE

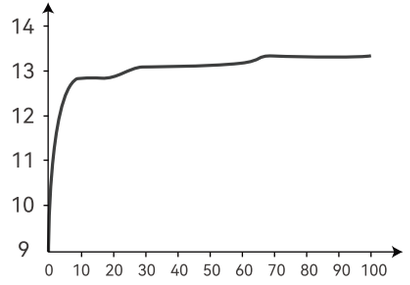
THE BATTERY CAPACITY

STATE OF CHARGE (SOC)

The battery capacity could be roughly estimated by its **resting voltage (notcharging/discharging voltage).**①

Since the voltage of each battery is slightly different, and the voltage measurement is affected by the measuring instrument, ambient temperature, etc., **the following parameters are for reference only.** The actual SOC of the battery is based on the discharge capacity under load.

Resting Voltage: The voltage is measured after the battery has been disconnected from the charger and loads with zero current, and left alone for 3 hours.



SOC(%)	VOLTAGE(V)
0	10 to 12
25	13 to 13.15
50	13.15 to 13.2
75	13.3 to 13.33
100	≥13.33②

① Based on the characteristics of LiFePO4 batteries, the voltage measured by all LiFePO4 batteries during charging/discharging is not the real voltage of the battery. Therefore, after charging/discharging and disconnecting the battery from the power source, the voltage of the battery will gradually drop/increase to its real voltage.

② After this battery is protected from overcharge, the tested battery voltage (not the real voltage) will be lower than the real voltage. To calculate the SOC (%), add 0.5V to 0.7V to the tested battery voltage.

SERIES / PARALLEL

CONNECTION

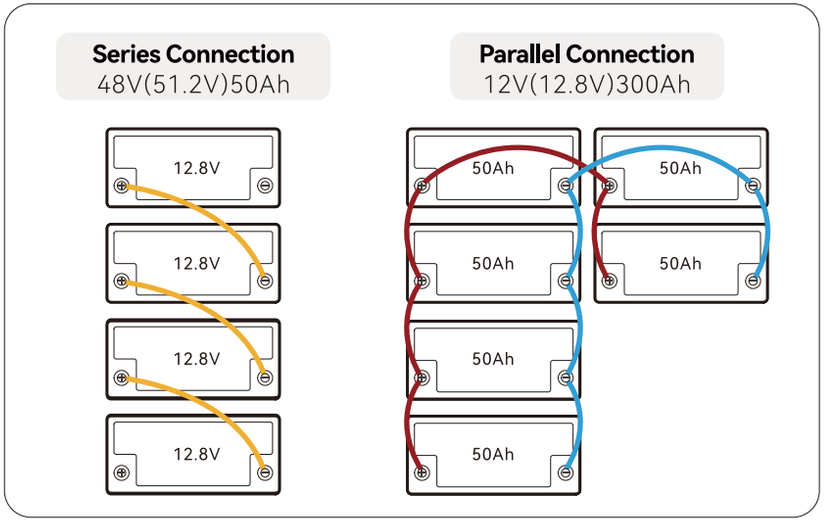
THE PREMISE OF CONNECTION

To connect in series or /and parallel, batteries should meet the below conditions:

a. All batteries must be identical, with the same capacity (Ah) and BMS rating (A).
b. They must be from the same brand, as different brands use different BMS designs.
c. They should be purchased around the same time (within one month of each other).
d. When using batteries in series or parallel, ensure their remaining capacities are close. It is recommended to fully charge each battery individually before series or parallel connection.
e. Series connection: Do not exceed 4 units in series. The voltage difference between any two units must be ≤1V.
f. Parallel connection: The voltage difference between any two units must be ≤0.5V.

LIMITATION FOR SERIES/PARALLEL CONNECTION

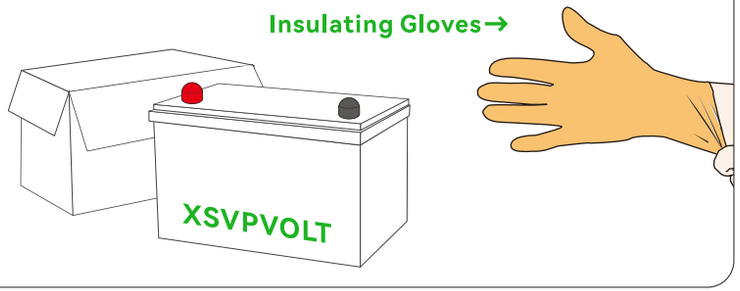
Supports connecting up to 24 identical batteries for up to 4S6P configuration:
4 in series for a **48V (51.2V)** battery system,
6 in parallel for a **300Ah** battery system,
or 4S6P (4 series × 6 parallel) for a **48V (51.2V) 300Ah** battery system.



HOW TO CONNECT BATTERIES

Step1 Wear Insulating Gloves

Wear Insulating Gloves for protection before connecting. Please payattention to operation safety in the process of connection.



Step2 Voltage Balancing Before Connection

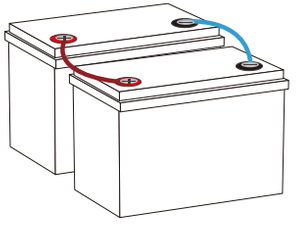
Below two steps are necessary to reduce the voltage difference between batteries and let the battery system perform the best of it in parallel.

Step①

Fully charge the batteries separately.
(voltage at rest: ≥13.33V)

Step②

Connect all of the batteries **in parallel**, and leave them together for **12~24hrs**.

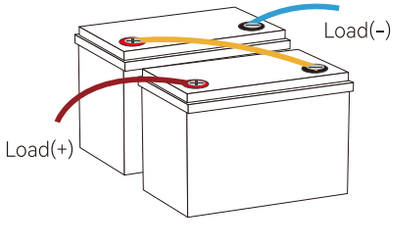


Step③

They're now ready for the **connection.**

Step3 Battery-to-Battery Connection

#1 Connect Batteries in Series ⊕ to ⊖

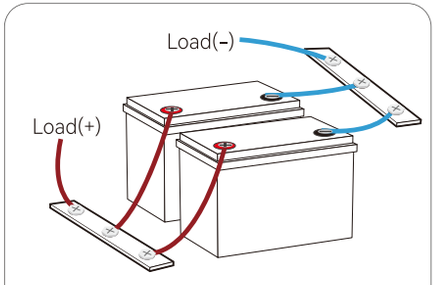


After series connection, the **voltage** of the battery system will be doubled according to the number of batteries you connect.

E.g. If two 12V 50Ah Smart Bluetooth TM batteries are connected in series, the battery system will be 24V (25.6V) 50Ah.

#2 Connect Batteries in Parallel ⊕ to ⊕ ⊖ to ⊖

Refer to Page 11 for total input & output connection



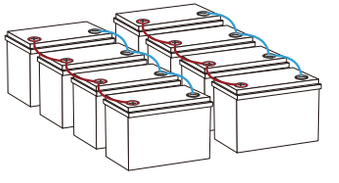
After parallel connection, the **capacity** of the battery system will be doubled according to the number of batteries you connect.

E.g. If two 12V 50Ah Smart Bluetooth TM batteries are connected in parallel, the battery system will be 12V (12.8V) 100Ah.

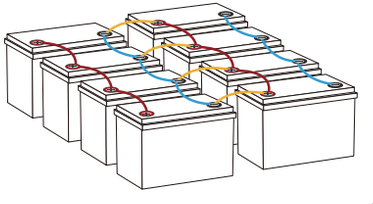
#3 Connect Batteries Both in Series & Parallel

Connect in parallel first, then series.

Step① Connect the batteries in **parallel.**



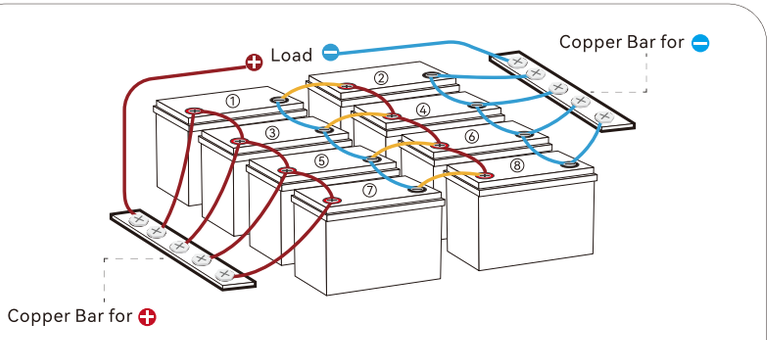
Step② Connect the paralleled battery systems in **series.**



Step4 Total Input & Output Connection

Use two **copper bars** (instead of battery terminals) to connect all the positive and negative output/input cables, ensuring that the input & output currents of each battery are balanced. (Not required when connecting batteries only in series.)

It is not recommended to use one terminal as the total positive or negative output/input of the battery system as the connected terminals may heat up or even melt if the total output/input current of the battery system is too high.



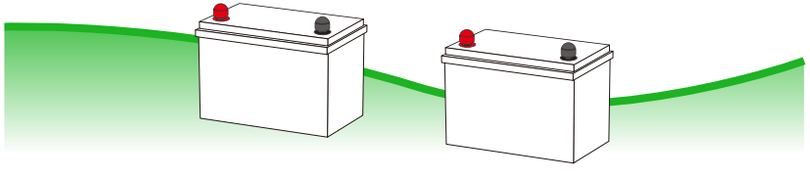
4P2S	Battery System	24V (25.6V)200Ah
	Energy	5,120Wh
	Max. Continuous Charge / Discharge Current	200A
	Max. Continuous Load Power	5,120W

① As ⊖ of ① / ③ / ⑤ / ⑦ is connected in series with ⊕ of ② / ④ / ⑥ / ⑧, please do not connect ⊖ of ① / ③ / ⑤ / ⑦ with ⊖ of load or ⊕ of ② / ④ / ⑥ / ⑧ with ⊕ of load, otherwise the battery system will fail to connect in series.

② Please do not connect in reverse order, which may affect the use of the batteries

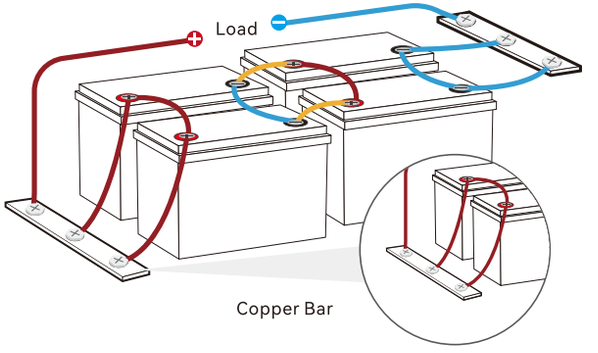
Step 5 Rebalancing Every 6 Months

It is recommended to rebalance the battery voltage every six months following Step 2 on Page 9 if you're connecting multiple batteries as a battery system, as there might be voltage differences after six months of the battery system running.

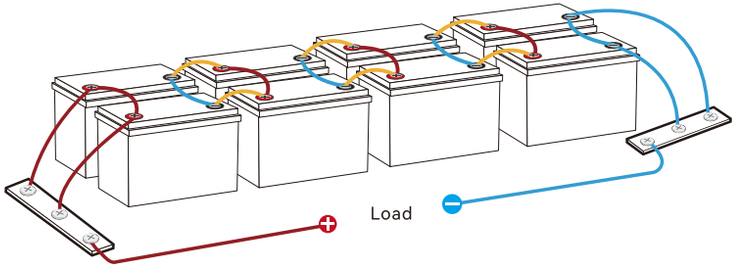


Wiring Diagram Reference

2P2S	Battery System	24V (25.6V)100Ah
	Energy	2,560Wh
	Max. Continuous Charge / Discharge Current	100A
	Max. Continuous Load Power	2,560W



2P4S	Battery System	48V (51.2V)100Ah
	Energy	5,120Wh
	Max. Continuous Charge / Discharge Current	100A
	Max. Continuous Load Power	5,120W



WHAT TO DO WHEN THE

BATTERY STOPS WORKING?

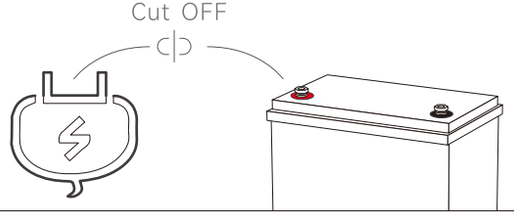
When the battery **① can't work; ② can't be charged; ③ voltage <10V**, it has 85% chances that BMS has shut it off for protection, and you could try one of below ways to activate the battery.

GENERAL STEPS

If the BMS has cut off the battery for protection, follow the below steps to activate it.

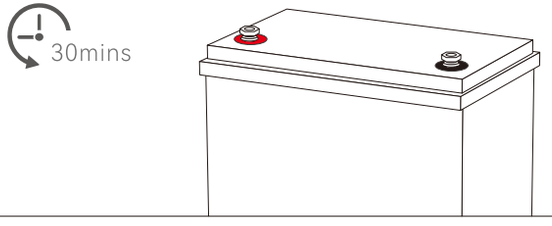
Step①

Cut off all the connections from the battery.



Step②

Leave the battery aside for 30mins
Then the battery will automatically recover itself to normal voltage (>10V) and can be used after fully charged.

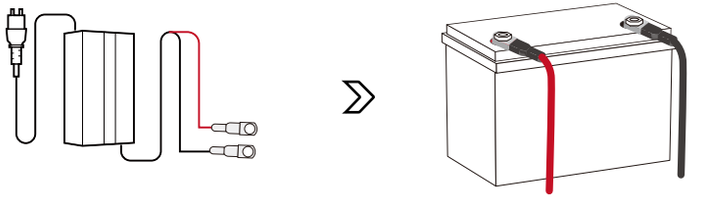


If the battery is unable to recover itself after the above steps, please try activating by ONE OF THE BELOW TWO METHODS.

After activated (voltage > 10V) and fully charged by the normal charging method, it can be used normally.

Method①

Use a **charger with lithium battery activation function** to fully charge the battery.

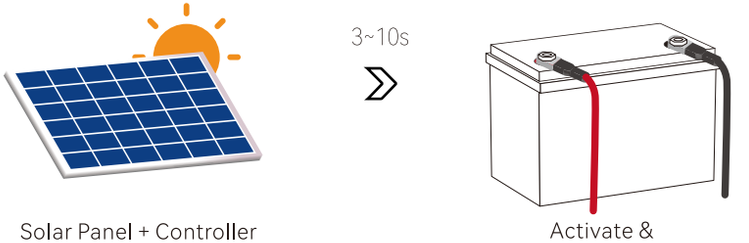


Charger with Lithium Battery Activation Function

Fully Charge

Method②

Connect **a controller** that supports 12V LiFePO4 battery charging to charge the battery for 3~10s in sunny daytime.



Solar Panel + Controller (Supports 12V LiFePO4)

Activate & Fully Charge

IMPORTANT SAFETY INSTRUCTION

1. Please keep the battery away from heat sources, sparks, flames, and hazardous chemicals.

2. Maintain adequate ventilation and heat dissipation.
Place the battery in a well-ventilated area with sufficient heat dissipation to prevent overheating and damage.

3. Size the battery cables and connectors appropriately.
Use high-stranded copper connectors and heavy gauge cables to handle possible battery loads. Make sure to keep identical cable lengths. Avoid accidents caused by unsuitable connectors or cables that make the connection a heat source during battery operation.

4. Please tighten all cable connections, as loose cable connections can cause terminal meltdown or fire.

5. DO NOT puncture, drop, crush, burn, penetrate, shake, or strike the battery.
The battery should be securely fastened during handling to prevent impact or dropping.
It should be safely secured to a solid plane, and the cables safely tied to a suitable location to avoid arcing and sparking due to friction.
DO NOT press it by placing heavy objects on top of it for long periods, which may damage it due to an internal short circuit.

6. DO NOT immerse the battery in water, whether in use or on standby.

7. DO NOT open, dismantle, or modify the battery.

8. DO NOT touch the exposed electrolyte or powder if the battery casing is damaged.

9. Exposed electrolyte or powder that has contacted the skin or eyes MUST be flushed out with plenty of clean water immediately. Seek medical attention afterward.

10. Avoid short circuit.
Please use circuit breakers, fuses, or disconnects that have been properly sized by certified electricians, licensed installers, or regional code authorities to protect all the electrical equipment in your system. The battery has a built-in battery management system (BMS) that protects the battery cells from over-charge, over-discharge, and over-current; however, this alone will not protect your system from severe electrical conditions.

11. Trained and certified technicians are required for safe and reliable installation. This product manual can only serve as a guideline as it cannot cover all possible scenarios.

12. Verify correct polarity.
Please verify the polarity before connecting the wiring. Reverse polarity can and will destroy the battery and other electrical equipment. Use a multimeter to determine proper polarity.

13. Avoid exposed metal terminals or connectors.
The terminals of this battery are always live. DO NOT place tools on the terminals, touch them with bare hands, short circuit them, or use them outside of specified electrical ratings.

14. DO NOT dispose of the battery as household waste. Please use recycling channels in accordance with local, state, and federal regulations.

WARNING

1. Batteries are potentially hazardous. Proper precautions must be taken during operation and maintenance.

2. Improper configuration, installation, or use of related equipment in the battery system may damage the battery and other components.

3. Wear appropriate personal protective equipment (PPE) when working on the battery.

4. Battery installation and maintenance must be performed by trained and certified technicians.

5. Failure to observe the above warnings may result in damage.

If you have any questions or require assistance, please contact us at <support-us@xsvpvolt.com> (please include your telephone number). We will respond within 24 hours.